

Owen Goodman

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EDUCATION

Virginia Tech <i>Master of Engineering in Computer Science and Applications</i>	Blacksburg, VA <i>Expected May 2027</i>
Virginia Tech <i>Bachelor of Engineering in Computer Science</i>	Blacksburg, VA <i>Expected May 2026</i>

EXPERIENCE

Undergraduate Researcher <i>Virginia Tech, Dr. Luke Achenie's Lab</i>	Aug 2025 – Present <i>Blacksburg, VA</i>
- Developed blind symbolic regression framework using Large Language Models to discover physical equations (e.g., Arrhenius, Antoine) directly from experimental data without prior knowledge of equation forms. - Implemented multi-stage compositional algorithm combining tree-based systematic exploration with LLM-guided hypothesis generation. - Deployed and benchmarked equation discovery pipelines on VT ARC supercomputing clusters using local vLLM servers and hybrid BFGS/differential evolution optimization.	

PROJECTS

CrateTests (SaaS) <i>Next.js, Firebase, OpenAI API, Stripe</i> - Developed an EdTech platform using fine-tuned GPT-4o-mini to generate study materials from user documents. - Integrated secure payment processing via Stripe and scaled to a paid user base at cratetests.com.
StructSprint <i>Next.js, React, Firebase, Vite</i> - Built a LeetCode-style competitive coding platform supporting timed problem solving, submissions, and real-time scoring. - Designed structured sprint workflows to simulate technical interview environments and reinforce data structures practice.
Voice Age & Gender Classifier <i>Python, PyTorch, ONNX, Flask, Librosa</i> - Engineered a Deep Learning system trained on the Mozilla Common Voice dataset for speaker classification. - Designed a CNN architecture for MFCC feature extraction, achieving 87% age and 99% gender accuracy. - Deployed model via ONNX and Flask to minimize inference latency for web-based audio processing.
2D Isometric Game Engine <i>Java, Swing, AWT</i> - Architected a custom 2D isometric rendering engine featuring optimized tile-based z-ordering. - Engineered an efficient pipeline managing 500+ active entities while maintaining a consistent 60 FPS.
Application Usage Monitor <i>C, Win32 API, JavaScript, Chart.js</i> - Built a Win32 system utility in C to track application persistence with high-resolution kernel timers. - Developed a local REST server to transmit JSON telemetry to a dashboard with interactive visualizations.

LEADERSHIP & ACTIVITIES

Fraternity Executive Board Member <i>Kappa Alpha Epsilon Eta Chapter</i>	Nov 2024 – Nov 2025 <i>Blacksburg, VA</i>
Directed logistics/budgets for a 50+ member organization and managed safety risk-mitigation protocols.	

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Swift, JavaScript, SQL, Rust, Go, GDScript
Tools: PyTorch, ONNX, Flask, React, Next.js, Firebase, Win32 API, Git, Linux, HPC/ARC

RELEVANT COURSEWORK

Data Structures & Algorithms · Systems · Computer Org · Engineering Stats · Applied Combinatorics · Software Design